Date

OL	T	H	W	R
Sunny	Hot	High	Weak	No
Sunny	Mild	High	Strong	No
Cloudy	Hot	High	Weak	yes
Rainy	Mild	High	Weak	yes
Rainy	Cool	Normal	Weak	yes
Rainy	Cool	Normal	Strong	No
Cloudy	Mild	Normal	Strong	yes
Sunny	Cool	Normal	Weak	No

For Rain:

Rain	Total
yes	4
No	4

$$H(S) = -\left(\frac{4}{8}\log_2\frac{4}{8}\right) - \left(\frac{4}{8}\log_2\frac{4}{8}\right)$$

$$= -\left(0.5\log_2 0.5\right) - \left(0.5\log_2 0.5\right)$$

$$= -(0.5(-1)) - (0.5(-1))$$

$$= 0.5 + 0.5$$

$$H(S) = 1$$

Date

For Outlook:

outlook	Total	yes	No	Entropy
Sunny	3		3	0.000
Cloudy	2	2		0.000
Rainy	3	2	1	

For Rainy:

$$= -\left(\frac{2}{3}\log_2\frac{2}{3}\right) - \left(\frac{1}{3}\log_2\frac{1}{3}\right)$$

$$= -(0.666(-0.58C)) - (0.333x - 1.58C)$$

$$= 0.390276 + 0.528138$$

$$= 0.9184$$

Weighted Entropy:

$$H(S|Rain) = \frac{3}{8}(0) + \frac{2}{8}(0) + \frac{3}{8}(0.9184)$$

$$H(S|Rain) = 0.3444$$

$$IG(Rain) = 1 - 0.3444$$

$$IG(Rain) = \boxed{0.655C}$$

For Temperature:

Temperature	Total	yes	No	Entropy
Hot	2	1	1	1
Mild	3	2	1	0.9184
Cool	3	1	2	0.9184

Hot:

$$= -\left(\frac{1}{2}\log_2\frac{1}{2}\right) - \left(\frac{1}{2}\log_2\frac{1}{2}\right)$$

$$= -(0.5\log_2(0.5)) - (0.5\log_2(0.5))$$

$$= -(0.5 \times -1) - (0.5 \times -1)$$

$$= 0.5 + 0.5$$

$$= 1$$

Mild:

$$= -\left(\frac{2}{3}\log_2\frac{2}{3}\right) - \left(\frac{1}{3}\log_2\frac{1}{3}\right)$$

$$= -(0.666\log_2(0.666)) - (0.333\log_2(0.333))$$

$$= -(0.666 \times -0.584) - (0.333 \times -1.584)$$

$$= 0.390276 + 0.528138$$

$$= 0.9184$$

ools

$$= -\left(\frac{1}{3}\log_2\frac{1}{3}\right) - \left(\frac{2}{3}\log_2\frac{2}{3}\right)$$

$$= -(0.333\log_2(0.333)) - (0.666\log_2(0.666))$$

$$= 0.528138 + 0.390276$$

$$= 0.9184$$

$$H(S|\text{Temperature}) = \frac{2}{8}(1) + \frac{3}{8}(0.9184) + \frac{3}{8}(0.9184)$$

$$= 0.25 + 0.3444 + 0.3444$$

$$= 0.9388$$

$$IG(\text{temp}) = 1 - 0.9388$$

$$= \boxed{0.0612}$$

For Humidity:

Humidity	Total	yes	No	Entropy
High	4	2	2	1
Normal	4	2	2	1

Entropy For High and Normal.

$$= -\left(\frac{2}{4}\log_2\frac{2}{4}\right) - \left(\frac{2}{4}\log_2\frac{2}{4}\right)$$

$$= -(0.5\log_2 0.5) - (0.5\log_2 0.5)$$

$$= 0.5 + 0.5 = 1$$

Date

Weighted Entropy.

$$H(S|H) = \frac{4}{8}(1) + \frac{4}{8}(1)$$

$$= 0.5 + 0.5$$

$$H(S|H) = 1$$

$$IG(\text{Humidity}) = 1 - 1 = \boxed{0}$$

For Wind:

	Total	yes	No	Entropy
Weak	5	3	2	
Strong	3	1	2	